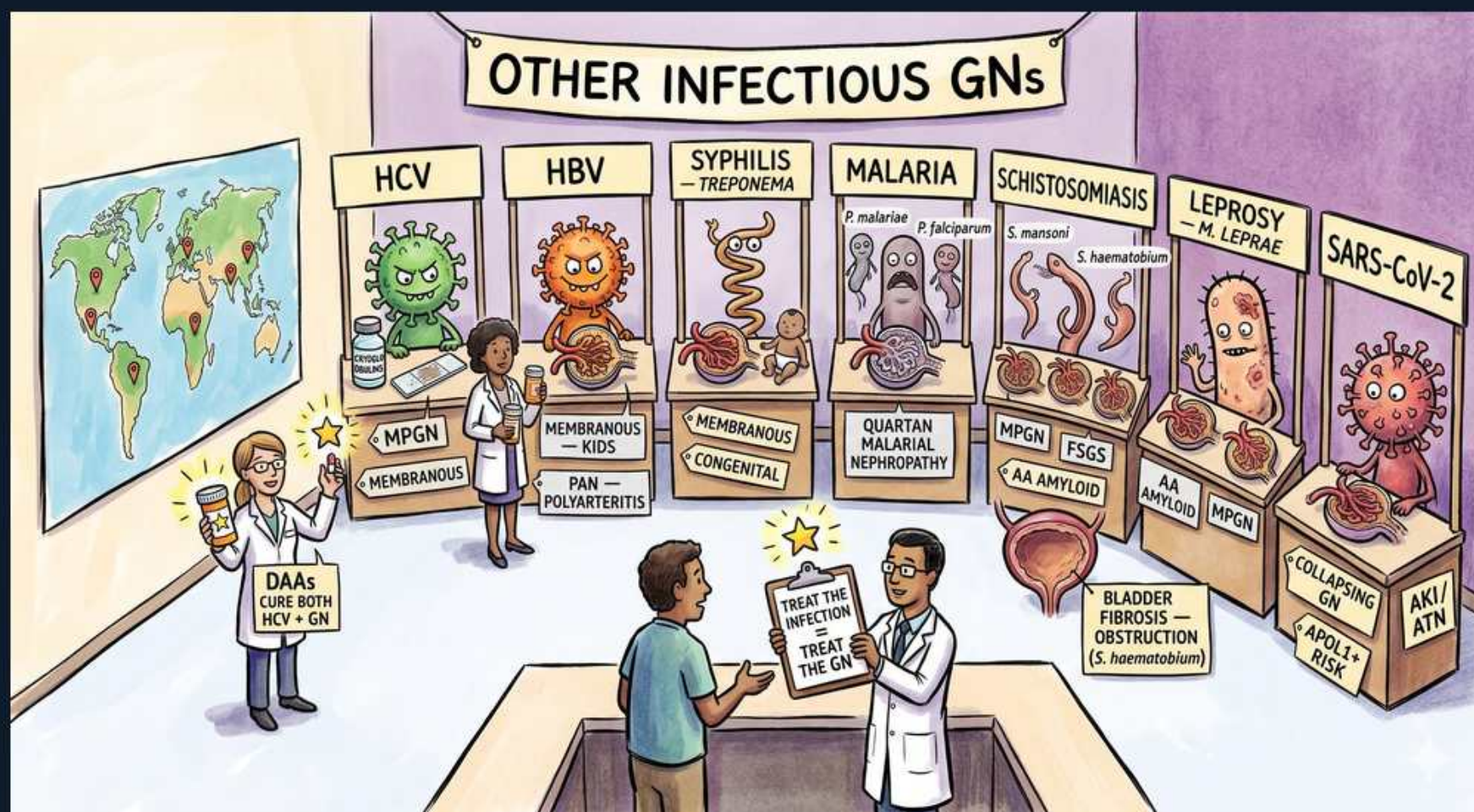


Glomerular (Nephrotic) — Infection-Related Nephropathies



1. Other infectious GNs banner

WHAT YOU SEE

Top banner OTHER INFECTIOUS GNs (museum of pathogen booths)

WHAT IT ENCODES

Many infections cause glomerular disease, and the histologic pattern is organism-specific: HBV→membranous, HCV→cryoglobulinemic MPGN, HIV→collapsing FSGS (HIVAN), syphilis→membranous, malaria→quartan nephropathy. The unifying board principle is that you treat the infection to treat the kidney rather than reflexively immunosuppressing. Complement level is the first triage step: post-strep GN, IE-GN, shunt nephritis, cryoglobulinemic MPGN and lupus run LOW complement (consume C3±C4), whereas IgA nephropathy, anti-GBM, ANCA-vasculitis and membranous run NORMAL complement.

BOARD TRAP

WRONG: treating these as primary/idiopathic glomerulonephritis and jumping to steroids. These are SECONDARY infection-associated GNs; the discriminator is a serologic/microbiologic clue (HBsAg, anti-HCV, blood cultures, HIV Ag/Ab) — find and treat the pathogen first, because immunosuppression alone can worsen an active infection.

2. HCV → MPGN/membranous

WHAT YOU SEE

HCV booth 'MPGN / membranous'

WHAT IT ENCODES

Hepatitis C is the classic cause of type I membranoproliferative GN (MPGN) driven by mixed (type II) cryoglobulinemia — IgM rheumatoid-factor complexes deposit in capillary loops giving a 'tram-track' double-contour GBM on PAS/silver stain. Presents with nephritic-nephrotic picture, palpable purpura, arthralgias, neuropathy and LOW C4 (C4 consumed more than C3). Cryoglobulins precipitate in the cold, so the sample must be kept warm. HCV can less commonly cause membranous nephropathy.

BOARD TRAP

WRONG: answering 'membranous' first for HCV. Discriminator: HCV's signature lesion is cryoglobulinemic MPGN with low C4, palpable purpura and a positive rheumatoid factor; reserve membranous as the HBV answer.

3. DAAs cure both HCV + GN

WHAT YOU SEE

Lower-left scientist 'DAAs cure both HCV + GN'

WHAT IT ENCODES

Direct-acting antivirals (e.g., sofosbuvir/velpatasvir, glecaprevir/pibrentasvir) achieving sustained virologic response are now first-line and frequently resolve the cryoglobulinemic MPGN by eliminating the antigenic drive. For severe/rapidly progressive cryoglobulinemic disease, add rituximab ± plasmapheresis and a short steroid course alongside antivirals. This is the prototype of 'treat the infection, cure the kidney.'

BOARD TRAP

WRONG: starting high-dose immunosuppression alone for HCV-cryo MPGN. Antiviral cure is the backbone; immunosuppression without antivirals risks increasing viral replication.

4. HBV → membranous (kids)

WHAT YOU SEE

HBV booth 'membranous — kids / PAN — polyarteritis'

WHAT IT ENCODES

Hepatitis B is the classic infectious cause of MEMBRANOUS nephropathy, especially in children (subepithelial immune deposits, 'spike-and-dome' on silver stain, often HBeAg-driven), and is the prototypical infectious trigger of polyarteritis nodosa (PAN — medium-vessel necrotizing vasculitis with renal microaneurysms on angiography, sparing the lungs and ANCA-negative). HBV-membranous in children often remits spontaneously with seroconversion; antivirals (entecavir, tenofovir) treat both the nephropathy and PAN.

BOARD TRAP

WRONG: attributing membranous to HCV or PAN to ANCA. Discriminators: HBV→membranous and HBV→PAN (ANCA-NEGATIVE, angiographic microaneurysms), whereas HCV→cryoglobulinemic MPGN. Do NOT use cyclophosphamide alone for HBV-PAN — pair it with an antiviral.

5. Syphilis → membranous, congenital

WHAT YOU SEE

Syphilis booth (Treponema) 'membranous / congenital'

WHAT IT ENCODES

Secondary and congenital syphilis (Treponema pallidum) can cause membranous nephropathy from immune-complex deposition; congenital syphilis is a recognized cause of nephrotic syndrome in the newborn. Renal disease typically resolves with penicillin treatment of the underlying infection. Confirm with treponemal (FTA-ABS/TP-PA) plus nontreponemal (RPR/VDRL) serology.

BOARD TRAP

WRONG: labeling pediatric/congenital membranous as idiopathic. Discriminator: a positive RPR/VDRL with reactive treponemal test points to syphilitic membranous — penicillin, not steroids, is the treatment.

6. Malaria → quartan nephropathy

WHAT YOU SEE

Malaria booth 'quartan malarial nephropathy'

WHAT IT ENCODES

Plasmodium malariae (the 'quartan' species, 72-hour fever cycle) causes chronic immune-complex 'quartan malarial nephropathy' — a steroid-resistant nephrotic syndrome/MPGN-like lesion most often in children in endemic regions. This contrasts sharply with P. falciparum, which causes ACUTE kidney injury, ATN and blackwater fever (massive intravascular hemolysis with hemoglobinuria).

BOARD TRAP

WRONG: blaming nephrotic syndrome on P. falciparum. Discriminator: chronic, steroid-RESISTANT nephrotic disease = P. malariae (quartan); acute AKI/blackwater hemoglobinuria = P. falciparum.

7. Schistosomiasis → MPGN/FSGS/AA

WHAT YOU SEE

Schistosomiasis booth 'MPGN / FSGS / AA amyloid' (S. mansoni / haematobium)

WHAT IT ENCODES

Schistosoma mansoni (and japonicum) cause glomerular disease — MPGN, FSGS and secondary AA amyloidosis from chronic egg-antigen immune complexes, often with hepatosplenic disease and portal hypertension. Diagnosis is by ova in stool/urine; praziquantel treats the infection. The species split is the high-yield point.

BOARD TRAP

WRONG: assigning glomerular disease to S. haematobium. Discriminator: S. mansoni/japonicum → liver + glomerulonephritis/AA amyloid, whereas S. haematobium → urinary tract (bladder).

8. Bladder fibrosis (S. haematobium)

WHAT YOU SEE

Center note 'BLADDER FIBROSIS / OBSTRUCTION (S. haematobium)'

WHAT IT ENCODES

Schistosoma haematobium lodges in the bladder venous plexus, producing terminal hematuria, chronic granulomatous cystitis, bladder wall fibrosis/calcification with obstructive uropathy, and is the classic risk factor for SQUAMOUS cell carcinoma of the bladder. Eggs have a terminal spine and are found in URINE. Praziquantel is treatment.

BOARD TRAP

WRONG: expecting glomerular disease or transitional-cell (urothelial) cancer from S. haematobium. Discriminators: haematobium → bladder obstruction and SQUAMOUS cell carcinoma (not urothelial), and it does NOT cause the glomerulonephritis seen with S. mansoni.

9. Leprosy → AA amyloid/MPGN

WHAT YOU SEE

Leprosy booth (M. leprae) 'AA amyloid / MPGN'

WHAT IT ENCODES

Chronic lepromatous leprosy (Mycobacterium leprae) is a sustained inflammatory state that drives serum amyloid A → secondary (AA) amyloidosis with nephrotic-range proteinuria, and can also produce MPGN. Any chronic infection (TB, osteomyelitis, bronchiectasis) shares this AA-amyloid pathway. Treat with multidrug therapy (dapsone, rifampin, clofazimine).

BOARD TRAP

WRONG: calling renal amyloid in a chronic infection 'AL' (primary/light-chain) amyloid. Discriminator: chronic infection/inflammation → AA amyloid (Congo-red apple-green birefringence, but fibrils are serum amyloid A, NOT immunoglobulin light chains); AL is the plasma-cell/myeloma type.

10. SARS-CoV-2 → collapsing FSGS

WHAT YOU SEE

SARS-CoV-2 booth 'collapsing GN / AKI / ATN'

WHAT IT ENCODES

COVID-19 can trigger collapsing glomerulopathy ('COVAN' — COVID-19-associated nephropathy), a collapsing variant of FSGS with massive proteinuria and AKI, occurring almost exclusively in patients carrying high-risk APOL1 genotypes (similar to HIVAN). The far more common renal manifestation, however, is ATN/AKI from the systemic illness. Same APOL1-collapsing biology links HIV, COVID and interferon therapy.

BOARD TRAP

WRONG: assuming every COVID AKI is collapsing FSGS. Discriminator: collapsing glomerulopathy occurs in APOL1 high-risk (typically African-ancestry) patients with nephrotic proteinuria, whereas the typical hospitalized COVID-AKI is ATN with bland sediment.

11. APOL1 risk

WHAT YOU SEE

Note 'APOL1 risk' under SARS-CoV-2

WHAT IT ENCODES

Two APOL1 risk alleles (G1/G2, enriched in West-African ancestry, evolved as protection against trypanosomiasis) markedly raise risk of collapsing glomerulopathy when a 'second hit' arrives — HIV (HIVAN), SARS-CoV-2 (COVAN), or interferon. APOL1 also drives a large share of non-diabetic FSGS and hypertension-attributed ESRD in this population. The unifying concept: viral/interferon trigger + APOL1 genotype = collapsing FSGS.

BOARD TRAP

WRONG: thinking HIVAN/COVAN is purely viral. Discriminator: the collapsing phenotype requires the APOL1 high-risk genotype as the host susceptibility factor; a viral trigger alone in an APOL1 low-risk patient typically does not produce collapsing FSGS.

12. AA amyloid (chronic infection)

WHAT YOU SEE

Repeated 'AA amyloid' tags (schisto, leprosy)

WHAT IT ENCODES

Chronic infections and chronic inflammation (leprosy, TB, schistosomiasis, osteomyelitis, bronchiectasis, inflammatory arthritis, familial Mediterranean fever) drive hepatic serum amyloid A production → secondary (AA) amyloidosis depositing in glomeruli as nephrotic-range proteinuria. Fat-pad or renal biopsy shows Congo-red apple-green birefringence; AA fibrils are derived from SAA protein. Controlling the underlying infection/inflammation is the key intervention.

BOARD TRAP

WRONG: choosing AL amyloid or congo-red alone as confirmatory of type. Discriminator: type the fibrils — chronic infection/inflammation = AA (serum amyloid A), monoclonal gammopathy/myeloma = AL (light chain). Both stain Congo red, so light-chain studies/immunohistochemistry distinguish them.

13. Treat the infection / treat the cure

WHAT YOU SEE

Center patient + doctor 'TREAT THE INFECTION / TREAT THE CURE'

WHAT IT ENCODES

The governing rule for all infection-related GN: eradicate the pathogen and the glomerular disease usually follows — DAAs for HCV-cryo MPGN, entecavir/tenofovir for HBV-membranous/PAN, ART for HIVAN, penicillin for syphilis, praziquantel for schistosomiasis, antibiotics/valve surgery for IE-GN. Post-streptococcal GN is the exception that is largely self-limited and supportive (BP/volume control). Add immunosuppression (steroids, rituximab, plasmapheresis) only for severe/rapidly progressive disease and alongside antimicrobial therapy.

BOARD TRAP

WRONG: empiric high-dose immunosuppression as monotherapy in infection-driven GN. Discriminator: unopposed immunosuppression can flare the underlying infection (e.g., HBV reactivation, worsening endocarditis) — antimicrobial therapy must anchor the plan.

14. World map / endemic distribution

WHAT YOU SEE

Left wall world map

WHAT IT ENCODES

Geography and exposure history reshape the differential: malaria and schistosomiasis (sub-Saharan Africa, tropics), leprosy (South Asia, Brazil), and APOL1-linked collapsing FSGS (West-African ancestry). A travel/exposure history (freshwater swimming for schisto, mosquito exposure for malaria) is often the discriminating clue on the boards.

BOARD TRAP

WRONG: omitting travel/exposure history when working up nephrotic syndrome. Discriminator: an endemic-region exposure (e.g., freshwater contact in Africa) reframes 'idiopathic' MPGN/FSGS as schistosomal or malarial and changes treatment to antiparasitics.

15. Post-strep GN (low complement, self-limited)

WHAT YOU SEE

Implied contrast — classic infection GN baseline

WHAT IT ENCODES

Post-streptococcal GN is the archetype infection-related GN to anchor against: nephritic syndrome 1-3 weeks after pharyngitis (or 3-6 weeks after impetigo) from group A strep, with LOW C3 (normalizes by 6-8 weeks), elevated ASO/anti-DNase B, and 'starry-sky' subepithelial humps on EM. Management is supportive (salt/water restriction, BP control); the prognosis is excellent in children. It is immune-complex mediated, not from active ongoing infection at the time of GN.

BOARD TRAP

WRONG: persistently low C3 beyond 8-12 weeks being called 'post-strep.' Discriminator: PSGN C3 normalizes within ~8 weeks — persistently low complement should redirect you to MPGN, C3 glomerulopathy, lupus, or endocarditis-associated GN.

16. IE-GN & shunt nephritis (low complement)

WHAT YOU SEE

Implied — endocarditis/shunt immune-complex GN

WHAT IT ENCODES

Infective endocarditis-associated GN and ventriculoatrial shunt nephritis are immune-complex GNs (often MPGN-like or crescentic) with LOW complement, frequently ANCA-positive (PR3) which is a classic mimic of primary vasculitis. Look for Staph aureus or viridans strep endocarditis with hematuria, RBC casts and a falling C3. Treatment is the antibiotics ± valve/shunt removal — the kidney recovers as the infection clears.

BOARD TRAP

WRONG: treating ANCA-positive crescentic GN with cyclophosphamide before excluding endocarditis. Discriminator: in IE-associated GN the ANCA is a red herring — blood cultures and an echocardiogram come first, and antibiotics (not immunosuppression) are the cure.